

MMatch: Semi-Supervised Discriminative Representation Learning for Multi-View Classification

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Abstract—Semi-supervised multi-view learning has been an important research topic due to its capability to exploit complementary information from unlabeled multi-view data. This work proposes MMatch, a new semi-supervised discriminative representation learning method for multi-view classification. Unlike existing multi-view representation learning methods that seldom consider the negative impact caused by particular views with unclear classification structures (weak discriminative views), MMatch jointly learns view-specific representations and class probabilities of training data. The representations concatenated to integrate multiple views' information to form a global representation. Moreover, MMatch performs the smoothness constraint on the class probabilities of the global representation to improve pseudo labels, whereas the pseudo labels regularize the structure of view-specific representations. A discriminative global representation is mined with the training process, and the negative impact of weak discriminative views is overcome. Besides, MMatch learns consistent classification while preserving diverse information from multiple views. Experiments on several multi-view datasets demonstrate the effectiveness of MMatch.

Index Terms—Semi-supervised learning, multi-view classification, discriminative representation, pseudo-labeling.

I. INTRODUCTION

REAL-WORLD data always exists in multi-view forms, such as voice, image, text, and so on [1], [2]. An example is that one image is described by text and color features [3], [4]. Additionally, there is a large amount of unlabeled

data within the real applications due to expensive labeling costs [5], [6]. An example is medical diagnosis data, which requires expensive machinery and manual labels by specialists. Hence, it is critical to explore intelligent methods to tackle real-world problems, learning representations from semi-supervised multi-view data. Multi-view data has two characteristics over a single view, i.e., consistency and complementarity. Consistency indicates that there are the same model's predictions between different views [7]–[9]. The complementary property means that each view has individual information, which helps the model combine different perspectives to make more accurate predictions [10]–[12]. Driven by these, numerous multi-view representation learning methods have been proposed and achieved significant success [13]–[15]. Joint representation methods [7], [16], [17] concatenate multiple views while they do not consider consensus property. Alignment representation methods [18], [19] maximize correlations for consistency, reducing the ability to preserve diversity. Moreover, discriminability is another characteristic among multiple views. Strong discriminative (clear classification structure) is beneficial to classification tasks (for example, it is easy to distinguish football and basketball from the semantic view, but it is problematic from the shape view). On the contrary, weak discriminative views hinder the effect of strong discriminative views and result in poor performance. However, minor works on multi-view discriminative learning were studied.

With the demands of making the most efficient use of labeled data, many semi-supervised multi-view learning methods have been proposed in the last decades, e.g., graph-based methods [7], [20] and regression-based methods [16], [21]. However, there are still limitations: first, graph-based methods are inefficient in dealing with out-of-sample data since they need to rerun the algorithms. Second, regression-based methods enforce regression loss as classification loss that usually causes incorrect penalization against the correct classification. Although many state-of-the-arts semi-supervised methods have been proposed recently [22]–[24], they are designed for single-view data.

To address these challenges, this work proposes MMatch, a novel semi-supervised discriminative representation learning method, as shown in Fig. 1. Each sample has two compact representations: a low-dimensional view-specific

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MMatch: 用于多视图分类的半监督判别式表示学习

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摘要——半监督多视图学习因其能够从无标签多视图数据中挖掘互补信息而成为重要的研究课题。本文提出了一种新的半监督判别式表示学习方法 MMatch，用于多视图分类。与现有多视图表示学习方法很少考虑分类结构不清晰的特定视图（弱判别视图）所带来的负面影响不同，MMatch 联合学习训练数据的视图特定表示和类别概率。这些表示被连接起来，整合多个视图的信息，形成全局表示。此外，MMatch 对全局表示的类别概率施加平滑约束以改进伪标签，而伪标签则对视图特定表示的结构进行正则化。通过训练过程挖掘出判别性全局表示，克服了弱判别视图的负面影响。此外，MMatch 在保留来自多个视图的多样化信息的同时，学习到一致的分类。在多个多视图数据集上的实验验证了 MMatch 的有效性。

索引词——半监督学习、多视图分类、判别式表示、伪标签。

由于标注成本高昂，实际应用中的数据面临诸多挑战[5][6]。例如，医疗诊断数据需要昂贵的设备和专家的人工标注。因此，探索智能方法来解决现实世界的问题至关重要，例如从半监督多视图数据中学习表征。多视图数据相对于单视图数据具有两个特征：一致性和互补性。一致性表明不同视图之间存在相同的模型预测[7] - [9]。互补性意味着每个视图都包含独特的信息，这有助于模型结合不同的视角做出更准确的预测[10] - [12]。基于这些特性，人们提出了许多多视图表征学习方法，并取得了显著的成功[13] - [15]。联合表征方法[7][16][17]将多个视图拼接在一起，但它们没有考虑一致性。对齐表征方法[18][19]最大化相关性以保证一致性，但这会降低保持多样性的能力。此外，可区分性是多视图数据的另一个特征。强判别性（清晰的分类结构）有利于分类任务（例如，从语义角度区分足球和篮球很容易，但从形状角度区分则比较困难）。相反，弱判别性视图会削弱强判别性视图的效果，导致性能下降。然而，关于多视图判别学习的研究还比较少。

为了更有效地利用标注数据，近几十年来提出了许多半监督多视图学习方法，例如基于图的方法[7]、[20]和基于回归的方法[16]、[21]。然而，这些方法仍然存在局限性：首先，基于图的方法处理样本外数据效率低下，因为它们需要重新运行算法。其次，基于回归的方法将回归损失作为分类损失，这通常会导致对正确分类的错误惩罚。尽管近年来提出了许多先进的半监督方法[22] - [24]，但它们都是针对单视图数据设计的。为了应对这些挑战，本文提出了一种名为 MMatch 的新型半监督判别式表示学习方法，如图 1 所示。每个样本都具有两个紧凑的表示：一个低维的、特定于视角的表示。

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现实世界的数据总是以多视图形式存在，例如语音、图像、文本等等[1][2]。例如，一幅图像可以用文本和颜色特征来描述[3][4]。此外，还有大量未标记的数据。

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representation by the encoder and a class probability by the classification head. The encoders and classification heads are view-specific, and no parameters are shared. To integrate complementary information from multiple views, MMatch concatenates view-specific representations to form a global representation. The unified pseudo labels are calculated on the global representation and refined through the enforced smoothness constraint. Specifically, pseudo labels aggregate information of nearby samples in the memory bank that MMatch stored some global representations and class probabilities to improve itself. These pseudo labels supervise view-specific discriminative representation learning. Therefore, each view can directly benefit from all views through the interaction of complementary information, mitigating the negative effects of weak discriminative views. As a result, MMatch achieves multi-view consistent classification while preserving complementary information among multiple views. Comprehensive experiments demonstrate the effectiveness of the proposed approach. We further provide insightful analyses about the consistent classification. In summary, the contributions of this work are as follows:

- This work proposes a novel semi-supervised discriminative multi-view representation learning framework, which achieves multi-view consistent classification while preserving complementary information among multiple views.
- The proposed method overcomes the negative impact caused by weak discriminative views and learns a strong discriminative global representation to perform multi-view classification.
- Discriminative representation learning and pseudo label generation interaction and promote each other, which provides another choice for semi-supervised multi-view learning. Experiments on different types of datasets demonstrate the effectiveness of classification performance.

II. RELATED WORK

A. Deep Semi-Supervised Learning

Semi-supervised learning (SSL) is a mature field with successful variant approaches [25]–[27]. Pseudo-labeling is a simple and effective SSL method, which trains models in a supervised way with labeled and unlabeled samples [28]. The corresponding method selects the maximum prediction probability as the pseudo labels of unlabeled samples. Pseudo-labeling results in the form of entropy minimization (EntMin) [29]. EntMin is an entropy regularization that encourages the model to generate low-entropy predictions for unlabeled samples. In theory, EntMin achieves decision boundary pass through low-density regions of data points region. Otherwise, it would be forced to assign low-confidence labels for unlabeled samples. Some methods [30], [31] utilize the sharpening function to encourage “soft” pseudo labels for entropy minimization. Whereas, FixMatch [23] achieves EntMin with threshold pseudo-labeling. It produces “hard” pseudo labels for unlabeled samples whose largest class probability is above a predefined threshold. Naive pseudo-label

learning depends on the class prediction quality of the model. When incorrect pseudo-label prediction occurs, mistakes will be accumulated, which is called confirmation bias. Therefore, recent SSL algorithms [23], [32] use pseudo-labeling as a component. CoMatch [32] provides a novel strategy to mitigate confirmation bias, imposing memory-smoothed pseudo-labeling in the embedding space. Inspired by the methods above, this work generates “hard” pseudo labels for entropy minimization and refined pseudo labels by aggregating information of neighboring samples in the feature representation space.

B. Multi-View Learning

1) *Multi-View Representation Learning*: Deep-learning-based multi-view representation learning methods can be roughly classified into three categories: joint representation, alignment representation, as well as share and specific representation [13], [33], [34]. Zhang *et al.* [35] extracted view-specific features with low-rank tensor constraint and learned the unified feature representation to integrate complementary information. Lan *et al.* [36] explores the complete information by performing consistency across different views and preserving diverse information of each view. Xu *et al.* [37] proposed a deep embedded collaborative training framework for multi-view clustering, which learns each view’s representation independently and iteratively sets one view as the referred view that guides the training of itself and others. Jia *et al.* [38] introduced a sharable and unique multi-view representation learning method. They designed two subnetworks for each view to extract specific and shared features with orthogonality and adversarial similarity constraints. However, these methods only exploit consistent and complementary information while ignoring the discriminability characteristic among multiple views. The proposed MMatch explores consistency and complementarity of multi-view and explores discriminability among multiple views to enhance classification performance.

2) *Semi-Supervised Multi-View Learning*: Semi-supervised multi-view learning has drawn much attention due to its practical value and promising performance. Correspondingly, a myriad of approaches have been proposed recently. These methods mostly follow the traditional SSL strategy and can be roughly divided into three categories. The first category is co-training-based methods [39]. They firstly use labeled data to train the classifiers and classify the unlabeled data of each view independently. After that, the predicted labels with high confidence in each view are chosen and added to the training set of another classifier. These methods repeat the procedure until they are converged. The second group is graph-based methods [7], [20] which are based on the manifold assumption that use labeled and unlabeled samples as vertices to construct a graph and propagation label information from labeled data to unlabeled data by edges. Nie *et al.* [7] proposed an auto-weighted graph-based multi-view learning approach that adaptively learns local structures and performs clustering or semi-supervised classification simultaneously. The third category is regression-based methods [21]. Instead of constructing the graph by directly calculating the similarity matrix, Yang *et al.* [21] constructed graphs by learning local

编码器表示数据点, 分类头表示类别概率。编码器和分类头均针对特定视图, 且不共享任何参数。为了整合来自多个视图的互补信息, MMatch 将特定视图的表示连接起来, 形成全局表示。统一的伪标签基于该全局表示计算, 并通过强制平滑约束进行优化。具体而言, 伪标签聚合了 MMatch 存储的内存库中邻近样本的信息, 该内存库存储了一些全局表示和类别概率, 用于自我改进。这些伪标签监督特定视图的判别性表示学习。因此, 每个视图都可以通过互补信息的交互直接受益于所有视图, 从而减轻弱判别性视图的负面影响。最终, MMatch 实现了多视图一致性分类, 同时保留了多个视图之间的互补信息。全面的实验证明了所提出方法的有效性。我们进一步对一致性分类进行了深入分析。总之, 本文的贡献如下:

- 本文提出了一种新颖的半监督判别式多视图表示学习框架, 该框架实现了多视图一致的分类, 同时保留了多个视图之间的互补信息。
- 所提出的方法克服了弱判别性视图带来的负面影响, 学习了强判别性全局表示以进行多视图分类。
- 判别式表征学习和伪标签生成相互促进、相互影响, 为半监督多视角学习提供了另一种选择。在不同类型数据集上的实验验证了该方法的分类性能。

二、RW

A. 深度半监督学习

半监督学习 (SSL) 是一个成熟的领域, 拥有众多成功的变体方法 [25] - [27]。伪标签是一种简单有效的半监督学习方法, 它利用已标记和未标记样本以监督的方式训练模型 [28]。该方法选择预测概率最大的伪标签作为未标记样本的伪标签。伪标签的结果以熵最小化 (EntMin) 的形式呈现 [29]。EntMin 是一种熵正则化方法, 它鼓励模型为未标记样本生成低熵预测。理论上, EntMin 能够使决策边界穿过数据点区域的低密度区域。否则, 模型将被迫为未标记样本分配低置信度的标签。一些方法 [30]、[31] 利用锐化函数来鼓励生成“软”伪标签以实现熵最小化。而 FixMatch [23] 则通过阈值伪标签来实现 EntMin。它为最大类别概率高于预定义阈值的未标记样本生成“硬”伪标签。

朴素伪标签学习依赖于模型的类别预测质量。当出现错误的伪标签预测时, 误差会不断累积, 这被称为确认偏差。因此, 近期的 SSL 算法 [23]、[32] 将伪标签作为其组成部分。CoMatch [32] 提出了一种新的策略来缓解确认偏差, 即在嵌入空间中引入记忆平滑的伪标签。受上述方法的启发, 本文通过在特征表示空间中聚合相邻样本的信息, 生成用于熵最小化的“硬”伪标签和精细化的伪标签。

B. 多视图学习

1) 多视图表征学习: 深度学习-

基于多视图表示学习的方法大致可以分为三类: 联合表示、对齐表示以及共享和特定表示 [13]、[33]、[34]。Zhang 等人 [35] 利用低秩张量约束提取视图特定特征, 并学习统一的特征表示以整合互补信息。Lan 等人 [36] 通过在不同视图之间保持一致性并保留每个视图的多样性信息来探索完整的信息。Xu 等人 [37] 提出了一种用于多视图聚类的深度嵌入式协同训练框架, 该框架独立学习每个视图的表示, 并迭代地将一个视图设置为参考视图, 以指导自身和其他视图的训练。Jia 等人 [38] 提出了一种可共享且独特的多视图表示学习方法。他们为每个视图设计了两个子网络, 以正交性和对抗相似性约束提取特定和共享特征。然而, 这些方法仅利用了一致性和互补性信息, 而忽略了多个视图之间的可区分性特征。所提出的 MMatch 探索多视图的一致性和互补性, 并探索多个视图之间的可区分性, 以提高分类性能。

2) 半监督多视图学习: 半监督多视图学习因其实用价值和良好的性能而备受关注。相应地, 近年来涌现出大量相关方法。这些方法大多遵循传统的半监督学习策略, 大致可分为三类。第一类是基于协同训练的方法 [39]。这类方法首先使用已标注数据训练分类器, 并独立地对每个视图的未标注数据进行分类。之后, 选择每个视图中置信度较高的预测标签, 并将其添加到另一个分类器的训练集中。这些方法重复上述过程直至收敛。第二类是基于图的方法 [7], [20]。这类方法基于流形假设, 将已标注和未标注样本作为顶点构建图, 并通过边将标签信息从已标注数据传播到未标注数据。Nie 等人 [21] 提出了一种基于图的半监督学习方法, 该方法利用图的边界条件, 通过图的边界条件将标签信息从已标注数据传播到未标注数据。[7] 提出了一种基于自加权图的多视图学习方法, 该方法能够自适应地学习局部结构并同时执行聚类或半监督分类。第三类是基于回归的方法 [21]。Yang 等人 [21] 并没有直接计算相似度矩阵来构建图, 而是通过学习局部结构来构建图。

linear regression models for each data and utilizing the label matrix as the unified regression target for all views to enhance consistency. *Tao et al.* [16] learned projection matrix for each view and weighted each view to evaluate the importance of different views. Besides, the out-of-sample data can be classified with the learned projection matrices and bias vectors. Unlike these methods, this work explores a pseudo-labeling SSL framework, which provides a new way of exploiting unlabeled multi-view data.

C. Memory-Based Learning

From a human perspective, there is a memory mechanism when we learn, and the memory mechanism can extract certain experience skills from memory to assist us in new learning when faced with new learning tasks. The method based on this mechanism in deep learning is called memory-based learning. Memory-based learning often introduces additional memory modules to store long-term memory. In this way, the model can more effectively use relevant experience for reasoning. *Zhang et al.* [40] proposed a model distillation method of deep mutual learning (DML) that enforced an ensemble of students to learn collaboratively and teach each other through the training process. In this work, we think that the student model of mutual teaching can also be regarded as a kind of external memory term. As a teacher model, the knowledge of the original setting is distilled to remember the key concepts of the final prediction. *He et al.* [41] present unsupervised representation learning through the proposed momentum contrast (MOCO). MOCO builds a large and consistent dictionary with a queue and a moving-averaged encoder for visual representation learning. In addition, different from the method of introducing an additional memory model, *Zheng et al.* [42] used the model itself as the memory model to align different predictions in the same domain for regularization training.

Different from the above methods, the proposed MMatch utilizes the moving-averaged memory bank as a component to refine pseudo-labels. According to the consistent characteristic of multiple views, MMatch performs consistent classification with the supervision of these pseudo-labels.

III. THE PROPOSED METHOD

A. Problem Statement

Given a multi-view dataset $\{x_i^1 \in \mathbb{R}^{d_1}, x_i^2 \in \mathbb{R}^{d_2}, \dots, x_i^V \in \mathbb{R}^{d_v}\}_{i=1}^N$. d_v is the dimensionality of v -th view, V is the number of views, and N represents the number of samples. Let $\mathcal{X}^v = \{x_i^v, y_i\}_{i=1}^B$ be a batch of labeled samples in v -th view, where x_i^v is the training sample, y_i is the corresponding label, and B is the batch size. Let $\mathcal{U}^v = \{u_i^v\}_{i=1}^B$ be a batch of unlabeled samples in the v -th view. Semi-supervised multi-view representation learning is to leverage semi-supervised multi-view data to learn a discriminative representation for the classification task.

B. The Proposed MMatch

Generally, there are multiple input forms and complementary information between multiple views. The proposed

MMatch trains the independent deep encoder network $f_{\theta^v}(\cdot)$ for each view to extract low-dimensional view-specific representations, where θ^v are the parameters. For each sample in the v -th view x_i^v , the corresponding representation can be described as:

$$z_i^v = f_{\theta^v}(x_i^v), \quad (1)$$

where $z_i^v \in \mathbb{R}^l$ is the latent representation of x_i^v in the l -dimensional feature space.

For each $f_{\theta^v}(\cdot)$, a classification head $h_{\phi^v}(\cdot)$ with ϕ^v parameters is constructed to predict the class probability based on the latent representation:

$$p_i^v = h_{\phi^v}(z_i^v), \quad (2)$$

where p_i^v is the class probability of x_i^v . Different views have different degrees of discriminability. The better the discriminability, the better the classification performance is. Thus, it is crucial to efficiently and effectively integrate complementary information from multiple views to improve discriminability. Therefore, we concatenate their latent representations to derive a global representation and train another global classification head $H_{\Phi}(\cdot)$ that predicts the unified class probability q .

$$Z_i = [z_i^1 z_i^2; \dots; z_i^v] \in \mathbb{R}^{\sum_{v=1}^V l}. \quad (3)$$

where “[]” indicates concatenation operation. Given a batch of labeled samples, the supervised loss $\mathcal{L}_x$ is optimized with the standard cross-entropy. Therefore, category-related features are saved as much as possible while the redundant features are reduced to clear the classification structure gradually [43]. The supervised loss of each view and the supervised loss of the global representation is described as:

$$\mathcal{L}_x^v = \frac{1}{B} \sum_{i=1}^B H(y_i, p_i^v), \quad (4)$$

$$\mathcal{L}_x^g = \frac{1}{B} \sum_{i=1}^B H(y_i, q_i), \quad (5)$$

where $\mathcal{L}_x^v$ and $\mathcal{L}_x^g$ are the supervised loss of v -th view and the supervised loss of global representation, respectively. MMatch explores unlabeled samples to train networks in a supervised way, taking pseudo labels on the global representations as “hard” labels for entropy minimization. These pseudo labels are confident where the global representations integrate complementary information of multiple views. However, pseudo-labeling methods always cause confirmation bias since they rely on the model’s predictions which the predictions mistakes would accumulate. To alleviate this issue, MMatch enforces smoothness constraint on the global representations space to improve pseudo labels.

For each sample in $\mathcal{X}^v$ and $\mathcal{U}^v$, its unified class probability q on the global representation Z can be obtained. It can be defined as the ground-truth $q = y$ and the model’s prediction $q = H_{\Phi}(Z)$ for the labeled and unlabeled sample, respectively. *Berthelot et al.* [22] proposed distribution alignment (DA) on unlabeled samples to facilitate the model to predict each class with equal probability. Following it, this work performs DA on our unlabeled samples: $q = DA(q)$. Specifically, we maintain

针对每组数据，我们构建了线性回归模型，并利用标签矩阵作为所有视图的统一回归目标，以增强模型的一致性。Tao 等人[16]学习了每个视图的投影矩阵，并对每个视图进行加权，以评估不同视图的重要性。此外，还可以利用学习到的投影矩阵和偏置向量对样本外数据进行分类。与这些方法不同，本文探索了一种伪标签 SSL 框架，为利用未标记的多视图数据提供了一种新途径。

C. 基于记忆的学习

从人类的角度来看，学习过程中存在一种记忆机制，该机制能够从记忆中提取某些经验技能，帮助我们应对新的学习任务。基于这种机制的深度学习方法被称为基于记忆的学习。基于记忆的学习通常会引入额外的记忆模块来存储长期记忆。这样，模型就能更有效地利用相关经验进行推理。Zhang 等人[40]提出了一种深度互学（DML）的模型蒸馏方法，该方法强制一组学生在训练过程中进行协作学习和互相教授。本文认为，互学的学生模型也可以被视为一种外部记忆项。作为教师模型，它蒸馏原始场景的知识，以记住最终预测的关键概念。He 等人[41]提出了一种基于动量对比（MOCO）的无监督表征学习方法。MOCO 利用队列和移动平均编码器构建了一个庞大且一致的字典，用于视觉表征学习。此外，与引入额外的记忆模型的方法不同，Zheng 等人 [42] 使用模型本身作为记忆模型，以对齐同一域中的不同预测进行正则化训练。与上述方法不同，本文提出的 MMatch 方法利用移动平均记忆库作为组件来细化伪标签。根据多视图的一致性特征，MMatch 在这些伪标签的监督下执行一致性分类。

三、TPM

A. 问题陈述

给定一个多视图数据集 $\{x \in \mathbb{R}, x \in \mathbb{R}, \dots, x \in \mathbb{R}\}$ ，其中 d 表示第 v 个视图的维度， V 表示视图的数量， N 表示样本的数量。令 $X=\{x, y\}$ 表示第 v 个视图中一批已标注的样本，其中 x 是训练样本， y 是对应的标签， B 是批次大小。令 $U=\{u\}$ 表示第 v 个视图中一批未标注的样本。半监督多视图表示学习旨在利用半监督多视图数据来学习用于分类任务的判别性表示。

B. 拟议的匹配

通常情况下，存在多种输入形式，且不同视图之间存在互补信息。所提出的

MMatch 为每个视图训练独立的深度编码器网络 $f(\cdot)$ ，以提取低维的视图特定表示，其中 θ 为参数。对于第 v 个视图 x 中的每个样本，其对应的表示可以描述为：

$$z = f(x), \quad (1)$$

其中 $z \in \mathbb{R}^l$ 是 x 在 l 维特征空间中的潜在表示。

对于每个 $f(\cdot)$ ，构建一个具有 ϕ 参数的分类头 $h(\cdot)$ ，以基于潜在表示预测类别概率：

$$p = h(z), \quad (2)$$

其中 p_i 表示 x 的类别概率。不同的视角具有不同的可区分性。可区分性越好，分类性能就越好。因此，高效整合来自多个视角的互补信息以提高可区分性至关重要。为此，我们将它们的潜在表示连接起来，得到一个全局表示，并训练另一个全局分类头 $H(\cdot)$ 来预测统一的类别概率 q 。

$$Z = [z_1; \dots; z_V] \in \mathbb{R}^{\sum_{v=1}^V l}. \quad (3)$$

其中“ $[\]$ ”表示连接操作。给定一批已标注的样本，监督损失函数 L_s 采用标准交叉熵进行优化。因此，类别相关的特征被尽可能地保留，而冗余特征则被减少，从而逐步清晰分类结构[43]。每个视图的监督损失和全局表示的监督损失描述如下：

$$L_v = \frac{1}{B} \sum_{i=1}^B H(y_i, p_i), \quad (4)$$

$$L_g = \frac{1}{B} \sum_{i=1}^B H(y_i, q_i) \quad (5)$$

其中， L_v 和 L_g 分别表示第 v 个视图的监督损失和全局表示的监督损失。MMatch 利用未标记样本以监督的方式训练网络，并将全局表示上的伪标签作为“硬”标签以最小化熵。这些伪标签能够准确反映全局表示整合了多个视图的互补信息。然而，伪标签方法总是会引入确认偏差，因为它们依赖于模型的预测，而预测误差会不断累积。为了缓解这个问题，MMatch 对全局表示空间施加平滑约束以改进伪标签。

对于 X 和 U 中的每个样本，都可以获得其在全局表示 Z 上的统一类别概率 q 。它可以定义为标记样本的真实类别 $q = y$ 和模型对标记样本的预测 $q = H(Z)$ 。Berthelot 等人 [22] 提出了对标记样本进行分布对齐（DA）的方法，以使模型能够以相等的概率预测每个类别。本文也对标记样本执行 DA： $q = DA(q)$ 。具体来说，我们保持

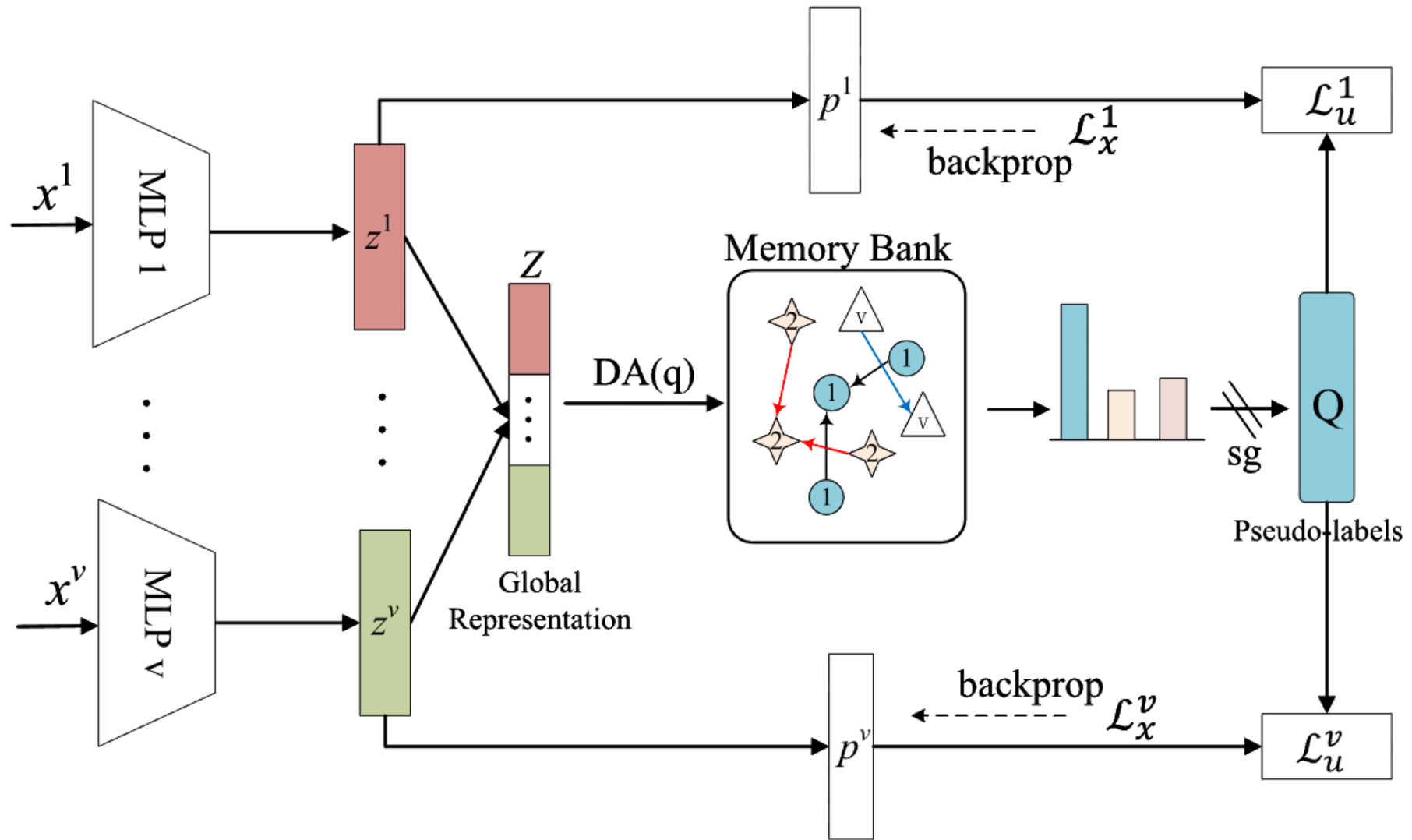


图 1. MMatch 的示意图。每个视图包含一个编码器和一个分类层。 z_i 是提取的视图特定表示， p_i 是分类分布。统一的伪标签 Q 经过细化，用于执行多视图判别式表示学习。“sg”表示停止梯度。

算法 1 MMatch: 用于多视图分类的半监督判别表示学习
输入: 多视图数据集 X 、 U ; X 的标签。

输出: 全局判别表示 Z 。

- 1 随机初始化 ϕ 和 θ 。
- 2 虽然未收敛
- 3 将嵌入式训练样本的每个原始视图视为 z ，并获取其类别概率 p ;
- 4 将每个视图的嵌入式表示连接起来，作为全局表示 Z 。
- 5 利用 (6) – (8) 预测未标记样本的标签。
- 6 使用 (10) 更新 MMatch 的参数 ϕ 和 θ 。
- 7 结尾

对 q 取移动平均值 $\tilde{q}$ ，并按比例 $q/\tilde{q}$ 对 q 进行缩放。给定缩放后的结果，我们对其进行重新归一化，以形成有效的概率分布: $\tilde{q} = \text{Normalize}(q/\tilde{q})$ ，其中 $\text{Normalize}(a) = a / \sum a$ 是的。

对于 X 和 U 中的每个样本，我们都能获得其全局特征 Z 。Li 等人 [32] 利用嵌入的结构来细化伪标签，从而缓解确认偏差。受此启发，MMatch 创建了一个记忆库来存储过去 M 个样本的类别概率和全局表示: $MB = \{q, Z\}$ (包括已标记和未标记的样本)。记忆库采用先进先出 (FIFO) 策略进行更新。对于当前未标记的样本，我们得到其全局表示 Z 和类别概率 q 。

精细化的伪标签 Q 可以从记忆库中具有相似嵌入表示的邻居中导出。因此，针对此目标优化的目标函数为:

$$J(Q) = (1 - \beta) \sum_{m=1}^M b_m Q - q + \beta Q - q, \quad (6)$$

其中，第一项鼓励当前样本 Q 与其邻近样本的类别概率相似。第二项用于保持自身相似性。 β 是第一项和第二项的平衡因子。在全局表示空间中， b 衡量当前样本与记忆库中第 m 个样本之间的相似性，其定义如下:

$$b = \frac{\exp(Z \cdot Z)}{\sum_{m=1}^M \exp(Z \cdot Z)}, \quad (7)$$

由于 b 已归一化 (例如, b 的总和为 1)，因此，最小值 $J(Q)$ 可推导如下:

$$Q = \beta q + (1 - \beta) \sum_{m=1}^M b q_m. \quad (8)$$

根据[23]，将值大于阈值 τ 的伪标签选为硬标签。每个视图的无监督分类损失 L_i 定义如下:

$$L = \frac{1}{B} \sum_{i=1}^B \mathbb{1}_{(\text{最大 } Q \geq \tau)} H(Q, p). \quad (9)$$

Overall, the training objective is defined as:

$$\mathcal{L} = \mathcal{L}_x^g + \sum_{v=1}^V (\mathcal{L}_x^v + \lambda \mathcal{L}_u^v), \quad (10)$$

where λ is the trade-off parameter. By optimizing $\mathcal{L}$, we can obtain these benefits as follows: 1) Each view can benefit from all views since the complementary information is interacted by pseudo labels. Therefore, the negative impact of weak discriminative views is mitigated. 2) MMatch performs a process: first, a small number of confidence pseudo labels are generated, then, the structure of global representation is gradually clear with training progress, lots of confident pseudo labels will be yielded, which are used to derive discriminative representations. 3) MMatch achieves consistent classification while preserving complementary information among multiple views.

C. Consistent Semi-Supervised Multi-View Classification

Theorem 1: Minimizing multiple cross-entropy with unified pseudo labels Q makes the class prediction probability p^v of multiple views tend to be consistent.

Proof. The optimization of multi-view classification loss of unlabeled data is:

$$\min \mathcal{L}_u^v = H(Q, p^v), v = \{1, 2, \dots, V\} \quad (11)$$

For any two views m and n from $v = \{1, 2, \dots, V\}$. These inequalities can be defined as:

$$H(Q, p^w) \geq H(Q), \quad i.e. \quad p^w \rightarrow Q \quad (12)$$

$$H(Q, p^n) \geq H(Q), \quad i.e. \quad p^n \rightarrow Q \quad (13)$$

where Q is fixed, thus, $p^w \rightarrow p^n$, this conclusion can be generalized to the case of multiple views.

Theorem 1 indicates that this work obtained a consistent class assignment with multiple views. It is impossible to get the same classification result of each view since the complementary between multiple views. Classify on the global representation mitigate the interference of a few wrong predictions caused by weak discriminative views. Therefore, the proposed approach achieves the definite classification predictions of higher confidence, and the final classification prediction is calculated by:

$$y = \operatorname{argmax}(h(Z)). \quad (14)$$

The proposed MMatch is illustrated in Fig. 1. Optimizing $\mathcal{L}$ only affects the encoders and classification heads. Thus, optimizing each view's embedded features and class distributions is independent of other views. As a result, the proposed framework can achieve the classification consistency of multiple views while preserving their feature's diversity. The algorithm is shown in Algorithms 1.

IV. EXPERIMENT SETUP

In this section, this work conducts several experiments to demonstrate the efficiency and effectiveness of the proposed MMatch on several real-world multi-view datasets. Firstly, this work introduces the datasets and the comparison methods.

A. Datasets

Here, this work selects five public datasets: HW, CUB, NUS-WIDE, Caltech101-7, WebKB. The details of these datasets are shown in Table I and the brief descriptions of all datasets are shown in the following:

- **HW** (Handwritten Numerals)¹ is a numeral dataset from 0 to 9, which contains 2000 samples with ten classes. Each class has 200 samples, and each sample has six features: 76 Fourier coefficients of the character shapes (FOU), 216 profile correlations (FAC), 64 Karhunen-love coefficients (KAR), 240-pixel averages in 2×3 windows (PIX), 47 Zernike moment (ZER) and 6 morphological (MOR) features.
- **CUB** (Caltech-UCSD)² contains 11788 bird images associated with text descriptions from 200 different classes. This work only uses 600 samples with two views that are 1024-dimensional features based on images with GoogLeNet and 300-dimensional features based on text.
- **NUS-WIDE** [44] is a real-world web image dataset of 25 classes. This work uses the first 120 images of each class, and five features are included: 64-dimensional color histogram, 144-dimensional color correlogram, 73-dimensional edge direction histogram, 128-dimensional wavelet texture, and 225-dimensional block-wise color moment.
- **Caltech101-7** [45] is a subset of images from Caltech. There are seven widely used classes with 1474 images selected: Motorbikes, Faces, Dollar-Bill, Garfield, Snoopy, Stop-Sign, and Windsor-Chair. Each image contains six views: 48 dimensions Gabor feature, 40 dimensions wavelet moments, 254 dimensions CENTRIST feature, 1984 dimensions HOG feature, 512 dimensions GIST feature, and 928 dimensions LBP feature.
- **WebKB** [46] contains 1051 web pages of 2 classes: 821 Non-Course pages and 230 Course pages. Each page has two views and these pages are collected from the computer science department websites at four universities: Cornell University, University of Washington, University of Wisconsin, and the University of Texas.

B. Comparison Methods

This work compares the proposed MMatch with several traditional and state-of-the-arts semi-supervised multi-view classification algorithms. The brief introductions of these methods are as follows:

- **AMGL** (Parameter-free auto-weighted multiple graph learning: a framework for multiview clustering and semi-supervised classification) [47]: one parameter-free auto-weighted multiple graph learning algorithm, it can be extended to multi-view clustering and semi-supervised classification.
- **MLAN** (Auto-Weighted Multi-View Learning for Image Clustering and Semi-Supervised Classification) [7]: one

¹<https://archive.ics.uci.edu/ml/datasets/Multiple+Features>

²<http://www.vision.caltech.edu/visipedia/CUB-200.html>

总体而言，培训目标定义为：

$$L = L + \sum_{v=1}^V (L + \lambda L), \quad (10)$$

其中 λ 为权衡参数。通过优化 L ，我们可以获得以下优势：1) 由于互补信息通过伪标签相互作用，每个视图都能从其他视图中受益。因此，弱判别性视图的负面影响得以减轻。2) MMatch 的工作原理如下：首先，生成少量置信度伪标签；然后，随着训练的进行，全局表示的结构逐渐清晰，产生大量置信度伪标签，这些伪标签用于导出判别性表示。3) MMatch 在保持多个视图之间互补信息的同时，实现了分类的一致性。

C. 一致的半监督多视图分类定理 1：最小化具有统一伪标签 Q 的多个交叉熵，使得多个视图的类别预测概率 p 趋于一致。

证明：针对无标签数据，多视角分类损失的优化为：

$$\min L = H(Q, p), v = \{1, 2, \dots, V\} \quad (11)$$

对于任意两个视图 m 和 n ，其中 $v = \{1, 2, \dots, V\}$ ，这些不等式可以定义如下：

$$\begin{aligned} H(Q, p) &\geq H(Q), \text{ 即 } p \rightarrow Q \quad (12) \\ H(Q, p) &\geq H(Q) \quad \text{IE } p \rightarrow Q \quad (13) \end{aligned}$$

当 Q 固定时，因此 $p \rightarrow p$ ，该结论可以推广到多视图的情况。

定理 1 表明，这项工作在多视角下获得了一致的类别分配结果。由于多视角之间的互补性，不可能每个视角都得到相同的分类结果。基于全局表示进行分类可以减轻由判别力较弱的视角导致的少数错误预测的干扰。因此，所提出的方法能够获得置信度更高的明确分类预测，最终分类预测的计算公式如下：

$$y = \operatorname{argmax}(h(Z)). \quad (14)$$

图 1 展示了所提出的 MMatch 算法。优化 L 仅影响编码器和分类头。因此，优化每个视图的嵌入特征和类别分布与其他视图无关。由此，所提出的框架能够在保持多个视图特征多样性的同时，实现多视图分类的一致性。

该算法如算法 1 所示。

IV. 实验部分 本节通过一系列实验，验证了所提出的 MMatch 算法在多个真实世界多视图数据集上的效率和有效性。首先，本文介绍了数据集和对比方法。

A. 数据集

本文选取了五个公开数据集：HW、CUB、NUS-WIDE、Caltech101-7 和 WebKB。这些数据集的详细信息见表 1，所有数据集的简要描述如下：

- HW（手写数字）是一个包含 0 到 9 的数字数据集，其中包含 2000 个样本，分为 10 个类别。每个类别有 200 个样本，每个样本有六个特征：76 个字符形状的傅里叶系数 (FOU)、216 个轮廓相关性 (FAC)、64 个 Karhunen-love 系数 (KAR)、 2×3 窗口中的 240 个像素平均值 (PIX)、47 个 Zernike 矩 (ZER) 和 6 个形态学 (MOR) 特征。
- CUB（加州理工学院-加州大学圣地亚哥分校）包含来自 200 个不同类别的 11788 张鸟类图像及其对应的文本描述。本研究仅使用了其中的 600 个样本，每个样本包含两种视图：一种是基于 GoogLeNet 提取的 1024 维图像特征，另一种是基于文本提取的 300 维特征。
- NUS-WIDE [44] 是一个包含 25 个类别的真实世界网络图像数据集。这项工作使用了每个类别的前 120 张图像，并包含了五个特征：64 维颜色直方图、144 维颜色相关图、73 维边缘方向直方图、128 维小波纹理和 225 维分块颜色矩。
- Caltech101-7 [45] 是加州理工学院图像集的一个子集。它选取了七个常用类别，共包含 1474 张图像：摩托车、人脸、美元钞票、加菲猫、史努比、停车标志和温莎椅。每张图像包含六个视图：48 维 Gabor 特征、40 维小波矩、254 维 CENTRIST 特征、1984 维 HOG 特征、512 维 GIST 特征和 928 维 LBP 特征。
- WebKB [46] 包含 1051 个网页，分为 2 类：821 个非课程网页和 230 个课程网页。每个网页都有两种视图，这些网页收集自四所大学的计算机科学系网站：康奈尔大学、华盛顿大学、威斯康星大学和德克萨斯大学。

B. 比较方法

本文将提出的 MMatch 算法与几种传统的和最先进的半监督多视图分类算法进行了比较。这些方法的简要介绍如下：

- AMGL（无参数自加权多图学习：多视图聚类 and 半监督分类的框架）[47]：一种无参数自加权多图学习算法，可以扩展到多视图聚类和半监督分类。
- MLAN（用于图像聚类和半监督分类的自动加权多视图学习）[7]：一

¹<https://archive.ics.uci.edu/ml/datasets/Multiple+Features>

²<http://www.vision.caltech.edu/visipedia/CUB-200.html>

TABLE I
OVERVIEW OF ADOPTED DATASETS

Dataset	No.of size	No.of views	No.of classes	No.of dimensionality for each view ($d_1, d_2, \dots, d_v$)
HW	2000	6	10	240/76/216/47/64/6
CUB	600	2	10	1024/300
NUS-WIDE	3000	5	25	64/225/144/73/128
Caltech101-7	1474	6	7	48/40/254/1984/512/928
WebKB	1051	2	2	2949/334

TABLE II
THE CLASSIFICATION ACCURACY (%) WITH DIFFERENT PERCENTAGES (σ) OF LABELED SAMPLES

Dataset	σ	AMGL	MVGL	MVAR	MLAN	LMVL	JCD	HLR-M ² VS	SPamCo	GRNet	Ours
HW	0.1	91.19	89.23	71.33	97.50	97.89	95.08	98.33	97.20	95.41	<u>98.05</u>
	0.2	93.78	89.31	96.35	97.75	97.95	95.94	99.31	97.43	97.03	<u>98.21</u>
	0.3	94.96	90.00	97.02	98.00	97.97	96.71	99.43	97.87	97.12	<u>98.39</u>
CUB	0.1	77.87	80.13	83.00	84.94	80.05	78.57	82.44	86.29	86.29	<u>86.04</u>
	0.2	83.70	81.94	85.62	87.72	85.85	86.50	<u>88.29</u>	87.08	88.12	89.26
	0.3	85.50	82.79	87.54	88.57	<u>90.78</u>	88.73	90.71	90.23	90.28	91.99
NUS-WIDE	0.1	16.39	19.49	13.41	23.69	8.22	22.37	21.75	23.74	23.74	24.47
	0.2	20.15	22.83	23.44	<u>27.30</u>	10.19	25.37	27.22	26.85	<u>28.43</u>	28.74
	0.3	22.72	24.79	27.46	29.30	13.83	27.82	<u>29.76</u>	29.31	29.71	32.77
Caltech101-7	0.1	87.80	14.29	86.03	93.53	74.81	91.96	93.76	93.83	<u>95.03</u>	95.54
	0.2	91.65	14.29	<u>96.03</u>	94.81	80.62	92.73	95.10	95.76	95.08	96.52
	0.3	93.58	14.29	<u>96.51</u>	96.20	80.77	94.57	92.57	<u>96.61</u>	96.02	97.33
WebKB	0.1	71.24	81.05	88.23	77.86	55.27	<u>93.34</u>	76.71	93.15	87.47	95.03
	0.2	72.52	82.16	92.00	77.77	71.66	94.64	87.72	<u>94.88</u>	92.90	96.55
	0.3	73.05	84.39	92.37	78.39	79.48	95.10	89.98	<u>95.51</u>	94.91	97.54

TABLE III
THE F1-SCORE (%) WITH DIFFERENT PERCENTAGES (σ) OF LABELED SAMPLES

Dataset	σ	AMGL	MVGL	MVAR	MLAN	LMVL	JCD	HLR-M ² VS	SPamCo	GRNet	Ours
HW	0.1	91.11	82.19	71.51	97.51	97.89	95.08	98.33	97.20	93.94	<u>97.97</u>
	0.2	93.74	85.27	96.35	97.78	97.94	95.94	99.31	97.43	96.47	<u>98.16</u>
	0.3	94.95	87.80	97.03	98.00	97.97	96.71	99.43	97.87	96.59	<u>98.29</u>
CUB	0.1	77.31	79.51	82.55	84.46	79.41	78.14	82.39	85.23	86.14	<u>85.47</u>
	0.2	83.43	81.24	86.00	87.60	85.38	86.37	86.54	86.84	88.11	<u>87.80</u>
	0.3	85.27	81.78	87.49	88.51	90.70	88.67	89.77	90.08	90.09	<u>90.65</u>
NUS-WIDE	0.1	16.23	18.22	13.14	<u>22.46</u>	6.24	21.89	20.76	22.64	21.37	21.16
	0.2	19.91	21.44	22.13	26.29	7.97	24.84	25.38	25.35	23.80	<u>26.07</u>
	0.3	22.36	23.10	25.51	<u>28.38</u>	12.17	27.21	27.56	28.03	26.01	28.64
Caltech101-7	0.1	52.27	10.05	55.35	71.39	37.49	70.74	<u>78.91</u>	73.68	77.52	79.81
	0.2	70.78	10.05	<u>84.19</u>	75.45	47.01	74.38	81.22	82.46	79.66	84.46
	0.3	78.25	10.05	<u>96.51</u>	96.20	80.77	94.57	73.48	84.81	79.72	97.33
WebKB	0.1	49.99	45.23	78.18	45.49	51.13	89.64	71.76	<u>90.78</u>	77.97	92.51
	0.2	50.17	45.72	87.25	47.11	66.37	91.61	83.40	<u>92.86</u>	88.26	93.47
	0.3	50.45	45.88	88.00	47.22	76.10	92.34	83.68	<u>93.62</u>	92.39	95.69

parameter-free method that automatically assigns the desired weight to each view. Besides, it can also learn local structure in semi-supervised classification.

- **MVAR** (Scalable multi-view semi-supervised classification via adaptive regression) [16]: a scalable regression-based multi-view semi-supervised method that is efficient in calculation and can solve large-scale semi-supervised classification problems. It automatically balances the

contributions of different views by adaptively optimized weight coefficients of each view.

- **MVGL** (Multi-view graph learning with adaptive label propagation) [20]: it constructs a robust and reliable graph from multi-view data with adaptive label propagation for semi-supervised classification. Besides, it constructs a common graph with the shared representation of multi-view data.

表一

OAD

Dataset	No.of size	No.of views	No.of classes	No.of dimensionality for each view ($d_1, d_2, \dots, d_v$)
HW	2000	6	10	240/76/216/47/64/6
CUB	600	2	10	1024/300
NUS-WIDE	3000	5	25	64/225/144/73/128
Caltech101-7	1474	6	7	48/40/254/1984/512/928
WebKB	1051	2	2	2949/334

表二 TCA (%) WDP (μ) LS

Dataset	σ	AMGL	MVGL	MVAR	MLAN	LMVL	JCD	HLR-M ² VS	SPamCo	GRNet	Ours
HW	0.1	91.19	89.23	71.33	97.50	97.89	95.08	98.33	97.20	95.41	<u>98.05</u>
	0.2	93.78	89.31	96.35	97.75	97.95	95.94	99.31	97.43	97.03	<u>98.21</u>
	0.3	94.96	90.00	97.02	98.00	97.97	96.71	99.43	97.87	97.12	<u>98.39</u>
CUB	0.1	77.87	80.13	83.00	84.94	80.05	78.57	82.44	86.29	86.29	<u>86.04</u>
	0.2	83.70	81.94	85.62	87.72	85.85	86.50	<u>88.29</u>	87.08	88.12	89.26
	0.3	85.50	82.79	87.54	88.57	<u>90.78</u>	88.73	90.71	90.23	90.28	91.99
NUS-WIDE	0.1	16.39	19.49	13.41	23.69	8.22	22.37	21.75	23.74	23.74	24.47
	0.2	20.15	22.83	23.44	<u>27.30</u>	10.19	25.37	27.22	26.85	<u>28.43</u>	28.74
	0.3	22.72	24.79	27.46	29.30	13.83	27.82	<u>29.76</u>	29.31	29.71	32.77
Caltech101-7	0.1	87.80	14.29	86.03	93.53	74.81	91.96	93.76	93.83	<u>95.03</u>	95.54
	0.2	91.65	14.29	<u>96.03</u>	94.81	80.62	92.73	95.10	95.76	95.08	96.52
	0.3	93.58	14.29	<u>96.51</u>	96.20	80.77	94.57	92.57	<u>96.61</u>	96.02	97.33
WebKB	0.1	71.24	81.05	88.23	77.86	55.27	<u>93.34</u>	76.71	93.15	87.47	95.03
	0.2	72.52	82.16	92.00	77.77	71.66	94.64	87.72	<u>94.88</u>	92.90	96.55
	0.3	73.05	84.39	92.37	78.39	79.48	95.10	89.98	<u>95.51</u>	94.91	97.54

表三

T F1-S (%) WDP (μ) LS

Dataset	σ	AMGL	MVGL	MVAR	MLAN	LMVL	JCD	HLR-M ² VS	SPamCo	GRNet	Ours
HW	0.1	91.11	82.19	71.51	97.51	97.89	95.08	98.33	97.20	93.94	<u>97.97</u>
	0.2	93.74	85.27	96.35	97.78	97.94	95.94	99.31	97.43	96.47	<u>98.16</u>
	0.3	94.95	87.80	97.03	98.00	97.97	96.71	99.43	97.87	96.59	<u>98.29</u>
CUB	0.1	77.31	79.51	82.55	84.46	79.41	78.14	82.39	85.23	86.14	<u>85.47</u>
	0.2	83.43	81.24	86.00	87.60	85.38	86.37	86.54	86.84	88.11	<u>87.80</u>
	0.3	85.27	81.78	87.49	88.51	90.70	88.67	89.77	90.08	90.09	<u>90.65</u>
NUS-WIDE	0.1	16.23	18.22	13.14	<u>22.46</u>	6.24	21.89	20.76	22.64	21.37	21.16
	0.2	19.91	21.44	22.13	26.29	7.97	24.84	25.38	25.35	23.80	<u>26.07</u>
	0.3	22.36	23.10	25.51	<u>28.38</u>	12.17	27.21	27.56	28.03	26.01	28.64
Caltech101-7	0.1	52.27	10.05	55.35	71.39	37.49	70.74	<u>78.91</u>	73.68	77.52	79.81
	0.2	70.78	10.05	<u>84.19</u>	75.45	47.01	74.38	81.22	82.46	79.66	84.46
	0.3	78.25	10.05	<u>96.51</u>	96.20	80.77	94.57	73.48	84.81	79.72	97.33
WebKB	0.1	49.99	45.23	78.18	45.49	51.13	89.64	71.76	<u>90.78</u>	77.97	92.51
	0.2	50.17	45.72	87.25	47.11	66.37	91.61	83.40	<u>92.86</u>	88.26	93.47
	0.3	50.45	45.88	88.00	47.22	76.10	92.34	83.68	<u>93.62</u>	92.39	95.69

这是一种无需参数的方法，能够自动为每个视图分配所需的权重。此外，它还能在半监督分类中学习局部结构。

- MVAR（基于自适应回归的可扩展多视图半监督分类）[16]：一种可扩展的基于回归的多视图半监督方法，计算效率高，能够解决大规模半监督分类问题。

它通过自适应优化每个视图的权重系数，自动平衡不同视图的贡献。

- MVGL（基于自适应标签传播的多视图图学习）[20]：它利用自适应标签传播技术，从多视图数据中构建鲁棒可靠的图，用于半监督分类。此外，它还构建了一个具有多视图数据共享表示的公共图。

- **LMSSC** (Latent Multi-view Semi-Supervised Classification) [48]: one method seeks the latent representation from original multiple views and performs graph construction, label prediction, and representation learning in a unified framework.
- **JCD** (Joint consensus and diversity for multi-view semi-supervised classification) [10]: learns complementary and diversity information of multiple views for semi-supervised multi-view classification, which is robust against certain low-quality views and boundary instances.
- **HLR-M²VS** (Hyper-Laplacian Regularized Multilinear Multiview Self-Representations for Clustering and Semisupervised Learning) [49]: one method performs hyper-Laplacian regularized multilinear multi-view representation learning. HLR-M²VS overcomes the negative impact caused by certain data features in multiple nonlinear subspaces and jointly learns global consistency among multiple views and view-specific local geometrical structures.
- **SPamCo** (Self-paced Multi-view Co-training) [50]: one co-training method designed for self-paced multi-view learning. It introduces hard and soft co-regularization terms to define different strategies for unlabeled data. Besides, SPamCo sets an extra hyper-parameter to reduce prediction bias resulting from diversity between different views. In the comparison experiment, this paper compares with the hard co-regularization strategies.
- **GRNet** (Graph-based remodeling network for multi-view semi-supervised classification) [51]: a graph-based method, utilizing graph regularization to leveraging unlabeled samples. Besides, GRNet can explore the potential relationship of multiple views and adaptively balances the consistency and complementary information of multiple views.

The implementations of AMGL and MLAN,³ MVGL,⁴ MVAR,⁵ LMVL,⁶ JCD,⁷ HLR-M²VS,⁸ and SPamCo⁹ are downloaded from the authors' websites.

C. Implementation Details

For each view, there are two components: An encoder with the structure Input—Fc512—Fc256—Fc256—Fc64 that extracts a low-dimensional representation and a classification head with the structure Fc64—Fc10. ReLu [52] and SGD [53] are chosen as the activate function and the optimizer with default learning rate 0.001 and weight decay 0.000005, respectively. The batchsize is set to 64. For the hyperparameters in MMatch that also exist in [32], we follow [32] and set $\tau = 0.95$, $\beta = 0.9$, and $M = 640$.

The comparison methods involved in this paper are downloaded from their author's websites, and the parameters are set

to follow the settings of each original paper. We take classification accuracy (ACC) and macro F1-score as the evaluation measures in the following experimental results presentations. A high value indicates a better classification performance for two metrics. Some comparison methods cannot give the predicted results on the test set. Thus, we use all data as the training set and evaluate the classification performance on the unlabeled data. All these results are recorded and averaged over five runs.

V. RESULTS AND ANALYSIS

In this section, we show the effectiveness and efficiency of the proposed approach from different aspects, and we give comprehensive analyses of these results.

A. Classification Results

This subsection evaluates the capability of the proposed MMatch with several representative semi-supervised multi-view methods. The best values are shown in bold black and the second-best values are underlined. The classification accuracy and the F1-score are shown in Table II and Table III, respectively. Through analyses of experiment results, these conclusions we can conclude: (1) From ACC and F1-score measure metrics, the proposed MMatch outperforms other methods across most datasets. Meanwhile, MMatch can perform well even if it does not achieve the best results on some datasets. Besides, we find there is a significant improvement on these datasets. For example, compared with the second-best values under the ACC metric, it achieves about 3% improvement on the NUS-WIDE dataset with $\sigma = 0.3$ and about 2% improvement on the WebKB dataset with $\sigma = 0.3$. Compared with the second-best value under the F1-score metric, it has about 1.73% over the second-best value on the WebKB dataset with $\sigma = 0.1$ and 0.9% improvement on the Caltech101-7 dataset with $\sigma = 0.1$.

(2) From the ACC metric, the performance of the proposed MMatch is more stable than compared with the comparison methods. For example, HLR-M²VS achieves the best values on HW, probably because the feature space of HW dataset is simple and the introduced hypergraph is easy to find the relationship between samples. However, it does not work well on WebKB dataset. MLAN has decent values on most datasets but poorly on WebKB. Besides, most comparing methods are limited to Caltech101-7 and WebKB datasets. The dimensionality of multiple views on the two datasets has large variation, i.e., are 48/40/254/1984/928 on Caltech101-7 and 2949/334 on WebKB. The discriminability of multiple views has significant gaps, and weak discriminative views cause a performance reduction in those semi-supervised methods. On the contrary, this work overcomes the negative impact of weak discriminative views. MMatch performs view-specific discriminative representation learning since each view directly benefits from complementary information with refined pseudo-labels.

(3) From the F1-score metric perspective, the graph-based methods reflect unsatisfactory classification performance on Caltech101-7 and WebKB datasets—especially AMGL and

³<http://www.esience.cn/peopole/fpnie/papers.html>

⁴<http://cobweb.cs.uga.edu/~shengli/publications.html>

⁵<https://github.com/taohong08>

⁶<https://github.com/sckangz/LMVL>

⁷<https://github.com/ChenpingHou>

⁸https://www.researchgate.net/profile/Yuan_xie4

⁹<https://github.com/Flowerfan/SPamCo>

- LMSSC (潜在多视图半监督分类) [48]: 一种方法从原始多视图中寻找潜在表示, 并在统一的框架中执行图构建、标签预测和表示学习。
- JCD (联合共识和多样性多视图半监督分类) [10]: 学习多个视图的互补和多样性信息, 用于半监督多视图分类, 对某些低质量视图和边界实例具有鲁棒性。
- HLR-MVS (超拉普拉斯正则化多线性多视图自表示聚类和半监督学习) [49]: 该方法执行超拉普拉斯正则化多线性多视图表示学习。HLR-MVS 克服了某些数据特征在多个非线性子空间中造成的负面影响, 并联合学习多个视图之间的全局一致性和视图特定的局部几何结构。
- SPamCo (自步多视图协同训练) [50]: 一种专为自步多视图学习而设计的协同训练方法。它引入了硬正则化和软正则化项, 为未标记数据定义了不同的策略。此外, SPamCo 还设置了一个额外的超参数, 以减少不同视图差异导致的预测偏差。在对比实验中, 本文将 SPamCo 与硬正则化策略进行了比较。
- GRNet (基于图的多视图半监督分类重构网络) [51]: 一种基于图的方法, 利用图正则化来利用未标记样本。此外, GRNet 可以探索多个视图之间的潜在关系, 并自适应地平衡多个视图的一致性和互补信息。

AMGL 和 MLAN、MVGL、MVAR、LMVL、JCD、HLR-M VS 和 SPamCo 的实现均从作者的网站下载。

C. 实现细节

每个视图包含两个组件: 一个结构为 Input—Fc512—Fc256—Fc256—Fc64 的编码器, 用于提取低维表示; 以及一个结构为 Fc64—Fc10 的分类头。激活函数和优化器分别选用 ReLU [52] 和 SGD [53], 默认学习率为 0.001, 权重衰减为 0.000005。批大小设置为 64。对于 [32] 中也存在的 MMatch 超参数, 我们沿用 [32] 的设置, 即 $\tau = 0.95$, $\beta = 0.9$, $M = 640$ 。本文所涉及的比较方法均从作者的网站下载, 参数也已设定。

为了遵循各篇原始论文的设置, 我们在以下实验结果展示中采用分类准确率 (ACC) 和宏 F1 分数作为评估指标。这两个指标的值越高, 表示分类性能越好。某些比较方法无法在测试集上给出预测结果。因此, 我们使用所有数据作为训练集, 并在未标记数据上评估分类性能。所有这些结果均记录并取五次运行的平均值。

V. RAA 在本节中, 我们将从不同方面展示所提出方法的有效性和效率, 并对这些结果进行全面分析。

A. 分类结果

本小节评估了所提出的 MMatch 算法在几种具有代表性的半监督多视图方法上的性能。最佳值以粗体黑色显示, 次佳值以下划线标出。分类准确率和 F1 分数分别列于表 II 和表 III。通过对实验结果的分析, 我们可以得出以下结论: (1) 从准确率 (ACC) 和 F1 分数这两个指标来看, 所提出的 MMatch 算法在大多数数据集上都优于其他方法。同时, 即使在某些数据集上未能取得最佳结果, MMatch 算法仍然表现良好。此外, 我们发现该算法在这些数据集上取得了显著的改进。例如, 与 ACC 指标下的次佳值相比, MMatch 算法在 $\mu = 0.3$ 的 NUS-WIDE 数据集上取得了约 3% 的改进, 在 $\mu = 0.3$ 的 WebKB 数据集上取得了约 2% 的改进。与 F1 分数指标下的第二好值相比, 它在 WebKB 数据集上比第二好值高出约 1.73% ($\mu = 0.1$), 在 Caltech101-7 数据集上比第二好值高出 0.9% ($\mu = 0.1$)。

(2) 从准确率 (ACC) 指标来看, 所提出的 MMatch 方法的性能比其他对比方法更加稳定。例如, HLR-M VS 在 HW 数据集上取得了最佳值, 这可能是因为 HW 数据集的特征空间较为简单, 且引入的超图易于找到样本之间的关系。然而, 它在 WebKB 数据集上的表现并不理想。MLAN 在大多数数据集上的表现尚可, 但在 WebKB 数据集上的表现较差。此外, 大多数对比方法仅限于 Caltech101-7 和 WebKB 数据集。这两个数据集上的多视图维度差异很大, Caltech101-7 数据集上的维度分别为 48/40/254/1984/928, 而 WebKB 数据集上的维度分别为 2949/334。多视图的区分能力存在显著差异, 弱区分度的视图会导致这些半监督方法的性能下降。相反, 本文克服了弱区分度视图带来的负面影响。MMatch 执行视图特定的判别性表示学习, 因为每个视图都直接受益于具有改进伪标签的互补信息。

(3) 从 F1 分数指标来看, 基于图的方法在 Caltech101-7 和 WebKB 数据集上的分类性能并不理想——尤其是在 AMGL 和

3 <http://www.escience.cn/peopole/fpnie/papers.html>

4 <http://cobweb.cs.uga.edu/~shengli/publications.html>

5 <https://github.com/taohong08>

6 <https://github.com/sckangz/LMVL>

7 <https://github.com/ChenpingHou>

8 https://www.researchgate.net/profile/Yuan_xie4

9 <https://github.com/Flowerfan/SPamCo>

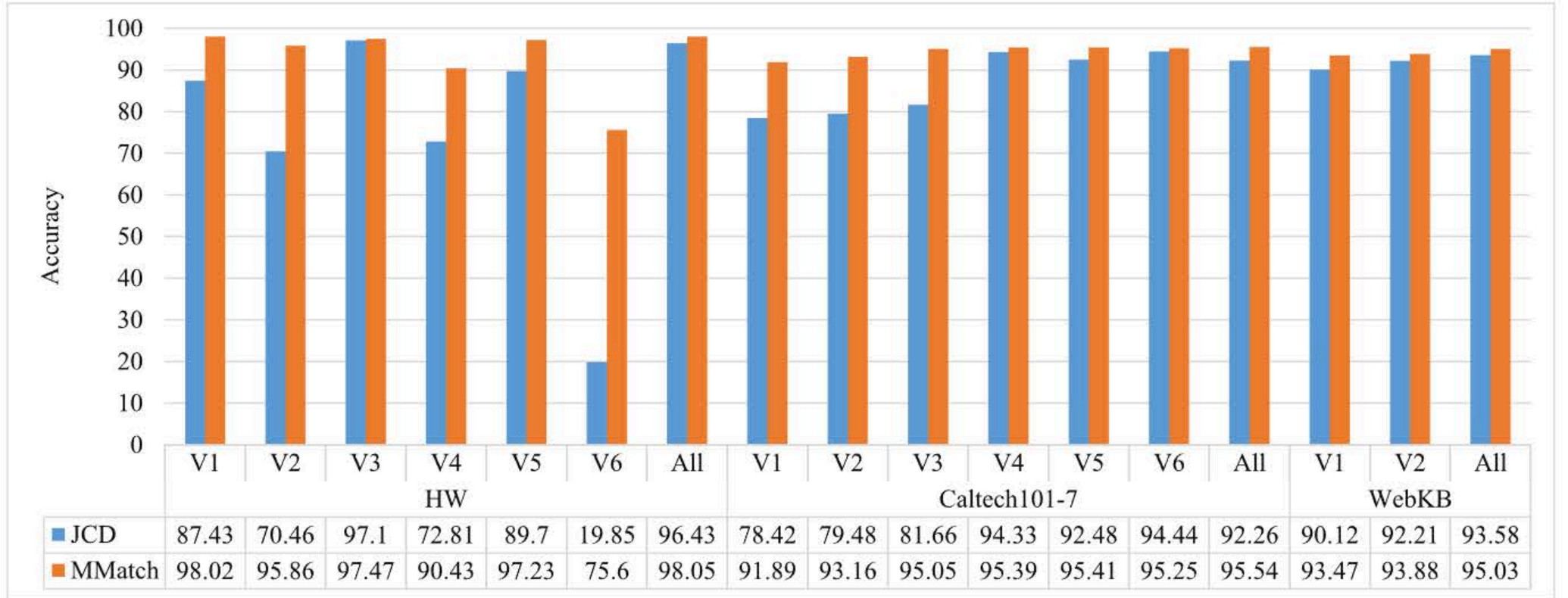


Fig. 2. We compare MMatch with JCD in each view. “V1,V2,...” represent the V-th view and “All” means all views.

MVGL. Caltech101-7 and WebKB datasets are extremely class unbalanced that the conventional graph-based methods are not suitable for them since they tend to choose balanced outcomes for the graph [54]. Yet even that, MMatch has achieved satisfactory classification performance. The reason is that MMatch produces pseudo-labels from a global perspective and refines them through the smoothness constraint on the global representation.

B. Discriminative of the Proposed MMatch

To evaluate the discriminability of the proposed MMatch, we compare the classification accuracy performance of each view with JCD [10]. From the JCD results shown in Fig. 2, we observed a significant difference in performance between multiple views, especially in the best view and the worst view. For example, about 77% of classification performance decreases between the “V3” and “V6” on HW dataset, and about 16% of classification performance decreases between “V1” and “V4” on Caltech101-7 dataset. The lower the classification performance, the less clear the classification structure is. In general, the classification performance of single-view is worse than multi-view. Still, the results of HW and Caltech101-7 are contrary that the classification result of multiple views is worse than single-view. The phenomenon indicates that particular weak discriminative views hinder the performance of highly discriminative views. Moreover, we observe that MMatch gains better performance than JCD in each view and reduces the performance gap among different views. The reason is MMatch mitigates the negative impact of weak discriminative views and learns view-specific discriminative representations. Besides, we also find that MMatch has a better performance on multi-view than the best view, which indicates that the preserved complementary information is helpful to boost classification performance. To prove more intuitively the high discrimination of the proposed MMatch, we use t-SNE [55] and UMAP [56] to visualize original views and embedded features by MMatch on HW, Caltech101-7, and

TABLE IV
ABLATION STUDY OF THE ENCODER NETWORK ($\sigma = 0.1$).
“AE” INDICATES THE ENCODER-DECODER NETWORK

Metrics	Method	HW	Caltech101-7	WebKB
ACC	ours	98.05	95.54	95.03
	AE	97.97	93.51	94.30
Precision	ours	98.11	81.75	94.04
	AE	98.12	75.53	88.67
Recall	ours	98.24	81.51	93.15
	AE	98.12	74.25	92.76
F1	Ours	97.97	79.81	92.51
	AE	97.92	72.13	90.23

WebKB with $\sigma = 0.1$. The results of the t-SNE visualization of the original views are shown in Fig. 3 (a) ~ (c). The results of the UMAP visualization of the original views are shown in Fig. 4 (a) ~ (c). The visual embedded features via these methods are shown in Fig. 3 (d) ~ (f) and Fig. 4 (d) ~ (f), respectively. We observe that original views are weak discriminative, especially on WebKB and Caltech101-7. On the contrary, the embedded features present more compact within classes and more considerable distances among classes, which is highly discriminative. The results further demonstrate that MMatch learns discriminative representations to boost classification performance.

C. Ablation Study

1) *Is the Decoder Necessary?*: The decoder network maps the latent representations back to the input, which plays an essential role in representation learning methods. Different from these methods, the encoder-only network is used in our model. So here, we investigate whether the decoder network is necessary for the proposed method. As shown in Table IV, the proposed MMatch achieves better performance than using the encoder-decoder network, which further empirically proves the rationale of using the encoder-only framework.

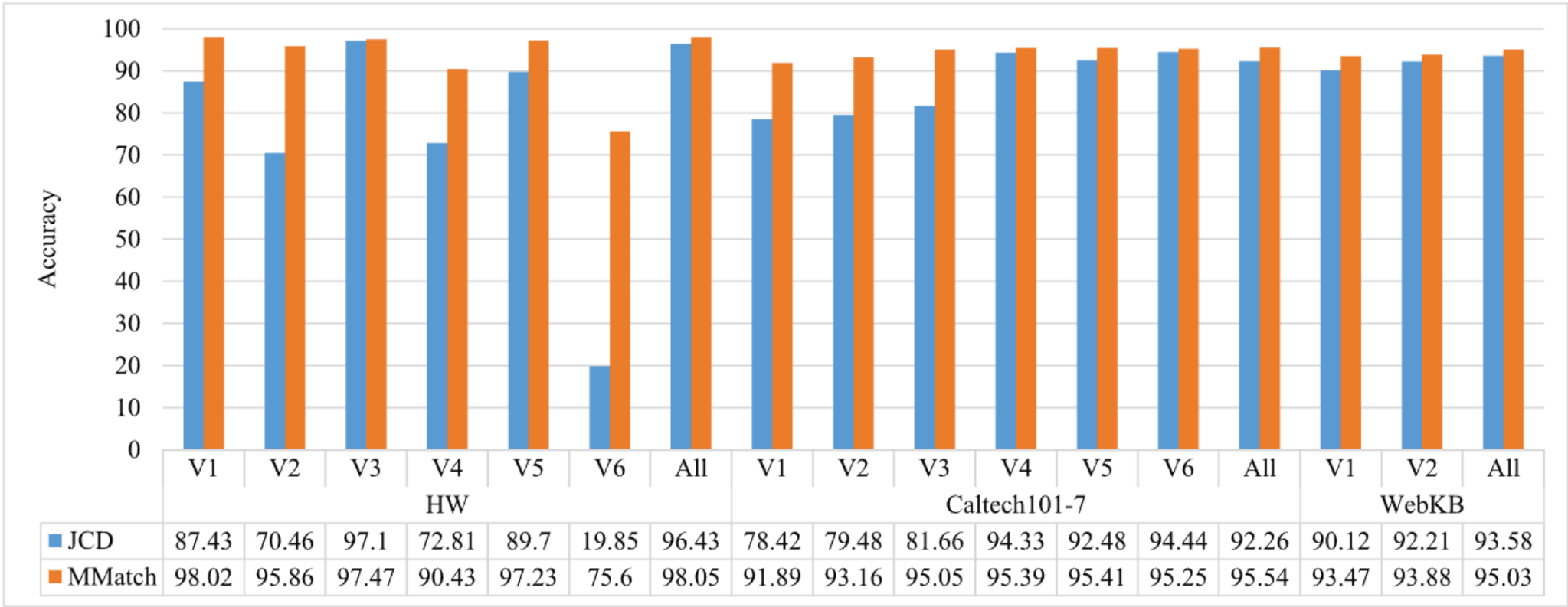


图 2。 我们从各个角度比较 MMatch 和 JCD。“V1,V2,...”代表第 V 个视图,“All”表示所有视图。

MVGL、Caltech101-7 和 WebKB 数据集的类别极度不平衡,传统的基于图的方法并不适用,因为它们倾向于选择图的平衡结果 [54]。然而,即便如此,MMatch 仍然取得了令人满意的分类性能。原因在于,MMatch 从全局角度生成伪标签,并通过全局表示的平滑性约束对其进行细化。

B. 拟议匹配的区分度

为了评估所提出的 MMatch 算法的区分能力,我们将每个视图的分类准确率与 JCD [10] 进行比较。从图 2 所示的 JCD 结果可以看出,不同视图之间的性能存在显著差异,尤其是在最佳视图和最差视图之间。例如,在 HW 数据集上,“V3”和“V6”之间的分类性能下降了约 77%,而在 Caltech101-7 数据集上,“V1”和“V4”之间的分类性能下降了约 16%。分类性能越低,分类结构越不清晰。一般来说,单视图的分类性能不如多视图。然而,HW 和 Caltech101-7 数据集的结果却相反,多视图的分类结果反而比单视图更差。这种现象表明,某些区分能力较弱的视图会阻碍区分能力较强的视图的性能。此外,我们观察到,MMatch 在每个视图上都比 JCD 表现更好,并且缩小了不同视图之间的性能差距。原因在于 MMatch 减轻了弱判别性视图的负面影响,并学习了特定于视图的判别性表示。此外,我们还发现 MMatch 在多视图上的性能优于最佳视图,这表明保留的互补信息有助于提升分类性能。为了更直观地证明所提出的 MMatch 的高判别性,我们使用 t-SNE [55] 和 UMAP [56] 对 HW、Caltech101-7 和 [56] 上的原始视图和 MMatch 嵌入的特征进行可视化。

表四
ASEN ($\mu = 0.1$)。
“AE” IE -DN

Metrics	Method	HW	Caltech101-7	WebKB
ACC	ours	98.05	95.54	95.03
	AE	97.97	93.51	94.30
Precision	ours	98.11	81.75	94.04
	AE	98.12	75.53	88.67
Recall	ours	98.24	81.51	93.15
	AE	98.12	74.25	92.76
F1	Ours	97.97	79.81	92.51
	AE	97.92	72.13	90.23

WebKB, $\mu = 0.1$ 。原始视图的 t-SNE 可视化结果如图 3 (a) ~ (c) 所示。原始视图的 UMAP 可视化结果如图 4 (a) ~ (c) 所示。通过这些方法提取的视觉嵌入特征分别如图 3 (d) ~ (f) 和图 4 (d) ~ (f) 所示。我们观察到,原始视图的区分能力较弱,尤其是在 WebKB 和 Caltech101-7 数据集上。相反,嵌入特征在类内呈现更紧凑的分布,在类间呈现更显著的距离,从而具有很强的区分能力。结果进一步表明,MMatch 学习到了具有区分性的表示,从而提升了分类性能。

C. 消融研究

1) 解码器是必要的吗? : 解码器网络图
将潜在表征还原为输入,这在表征学习方法中起着至关重要的作用。与这些方法不同,我们的模型仅使用编码器网络。因此,我们在此研究解码器网络对于所提出的方法是否必要。如表 IV 所示,所提出的 MMatch 模型比使用编码器-解码器网络取得了更好的性能,这进一步从经验上证明了使用仅编码器框架的合理性。

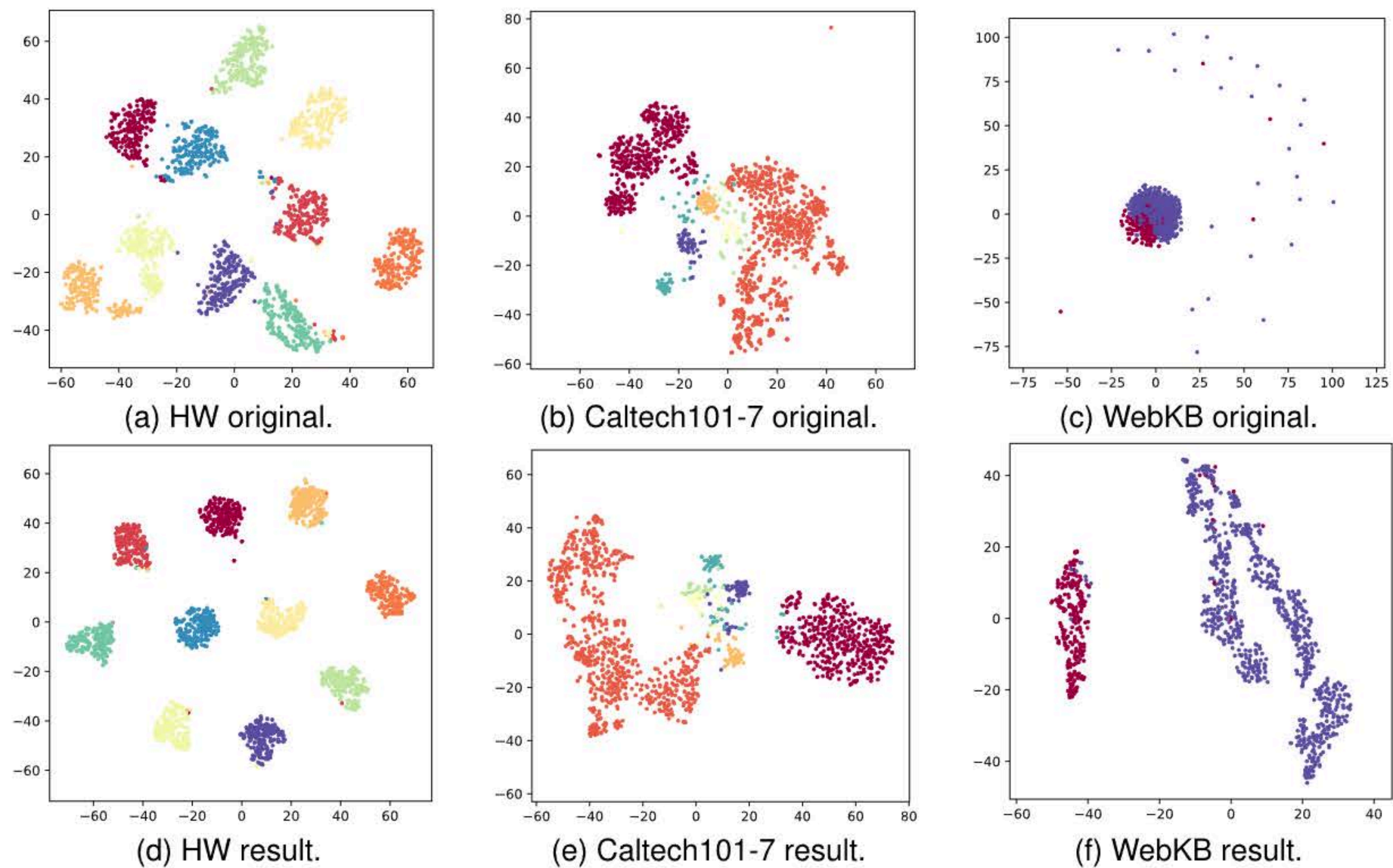


Fig. 3. t-SNE visualization of original datasets and representations after MMatch mapping.

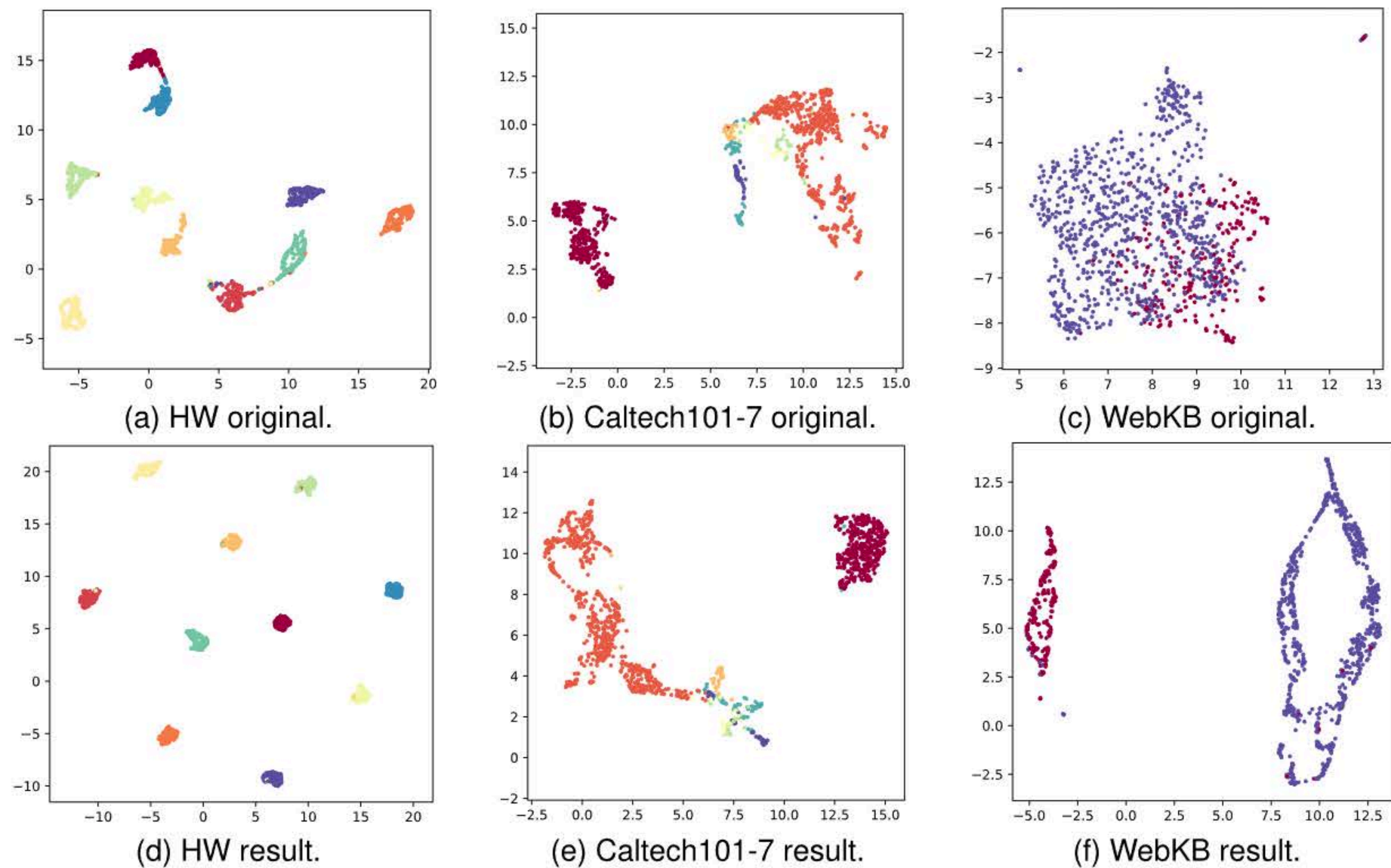


Fig. 4. UMAP visualization of original datasets and representations after MMatch mapping.

2) *Are the Multiple Independent Networks Effective?*: The purpose of multiple independent networks is to learn low-level view-specific representations. We compare it with a single

model (concatenate multiple views as a single view) training strategy to demonstrate its effectiveness. As shown in Table V, it is observed that the performance of multiple independent

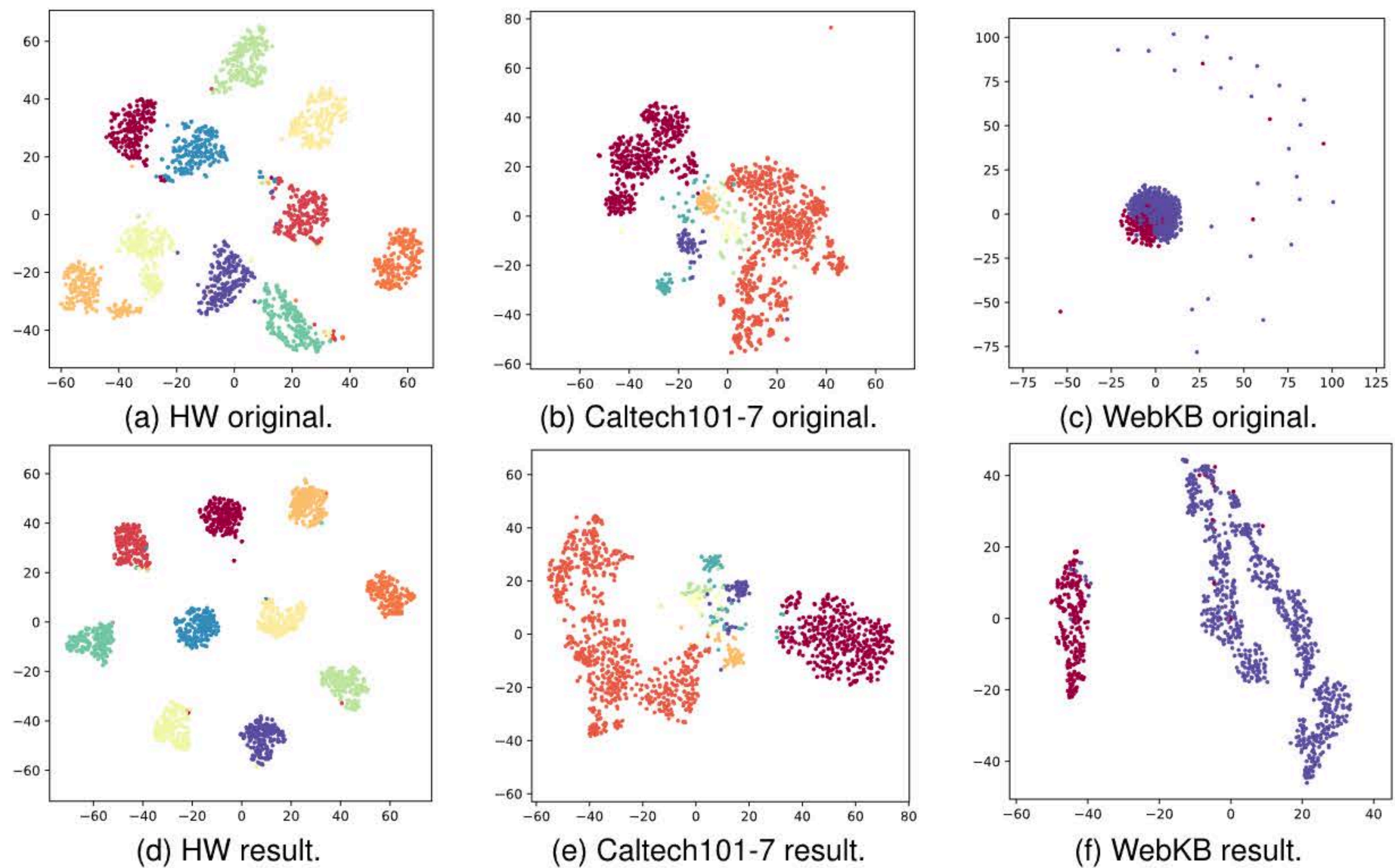


图 3. t-SNE 可视化原始数据集和 MMatch 映射后的表示。

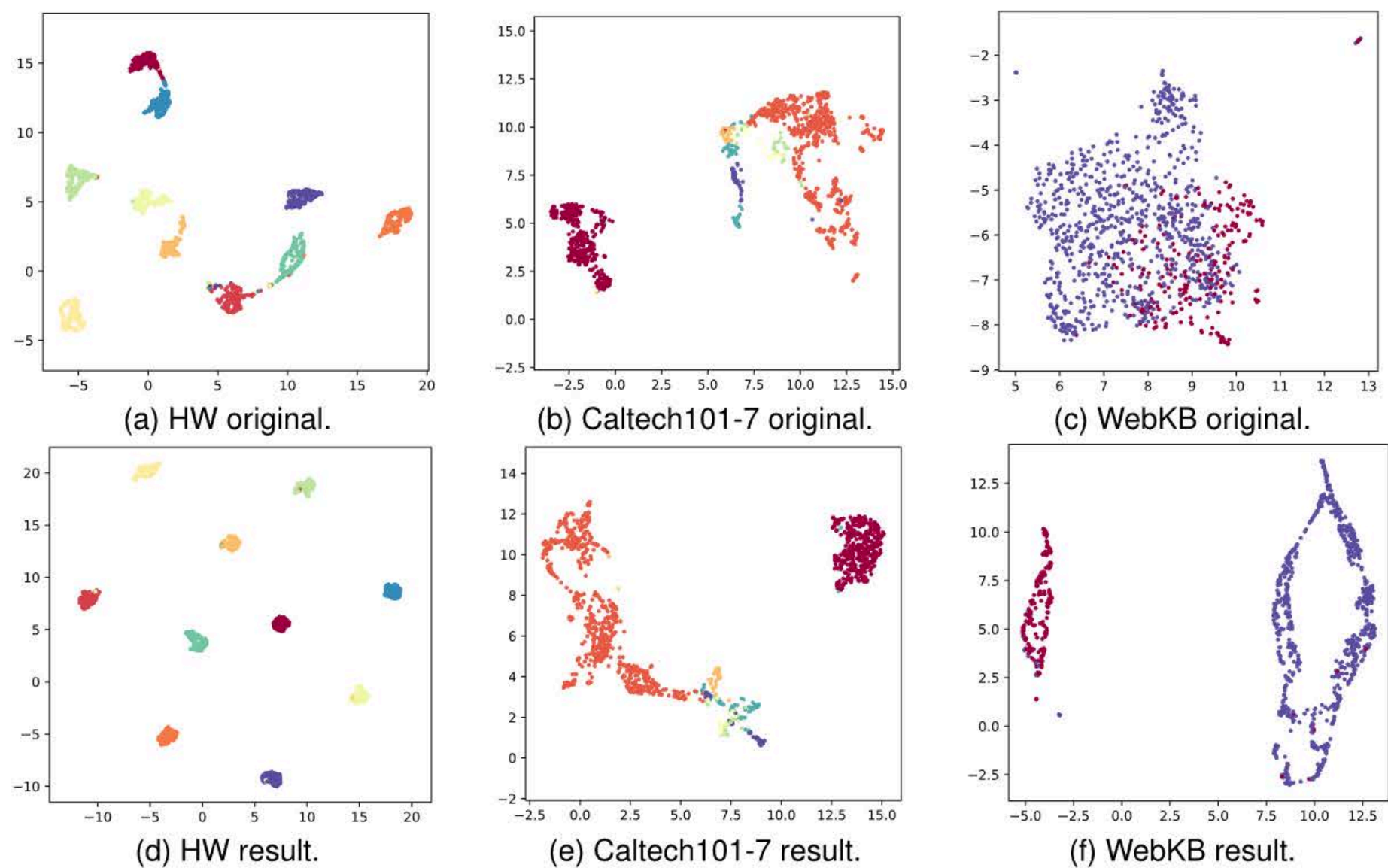


图 4. UMAP 可视化原始数据集和 MMatch 映射后的表示。

2) 多个独立网络是否有效?
多个独立网络的目的是学习底层特定于视图的表示。

我们将其与单一模型（将多个视图拼接成一个视图）训练策略进行比较，以证明其有效性。如表 V 所示，可以观察到多个独立模型的性能。

TABLE V

ABLATION STUDY OF MULTIPLE INDEPENDENT NETWORKS ($\sigma = 0.1$).
“SM” INDICATES A SINGLE MODEL

Metrics	Method	HW	Caltech101-7	WebKB
ACC	ours	98.05	95.54	95.03
	SM	94.64	92.80	89.83
Precision	ours	98.11	81.75	94.04
	SM	94.96	69.30	92.44
Recall	ours	98.24	81.51	93.15
	SM	94.72	65.07	76.60
F1	ours	97.97	79.81	92.51
	SM	94.25	65.05	80.81

TABLE VI

ABLATION STUDY OF MEMORY BANK ($\sigma = 0.1$). ‘✓’ AND ‘✗’ INDICATE
WITH/WITHOUT MEMORY BANK, RESPECTIVELY

Metrics	Method	HW	Caltech101-7	WebKB
ACC	✓	98.05	95.54	95.03
	✗	97.77	94.14	94.89
Precision	✓	98.11	81.57	94.04
	✗	97.99	71.41	90.50
Recall	✓	98.24	81.51	93.51
	✗	97.79	69.76	79.20
F1	✓	97.97	79.81	92.51
	✗	97.65	68.50	80.72

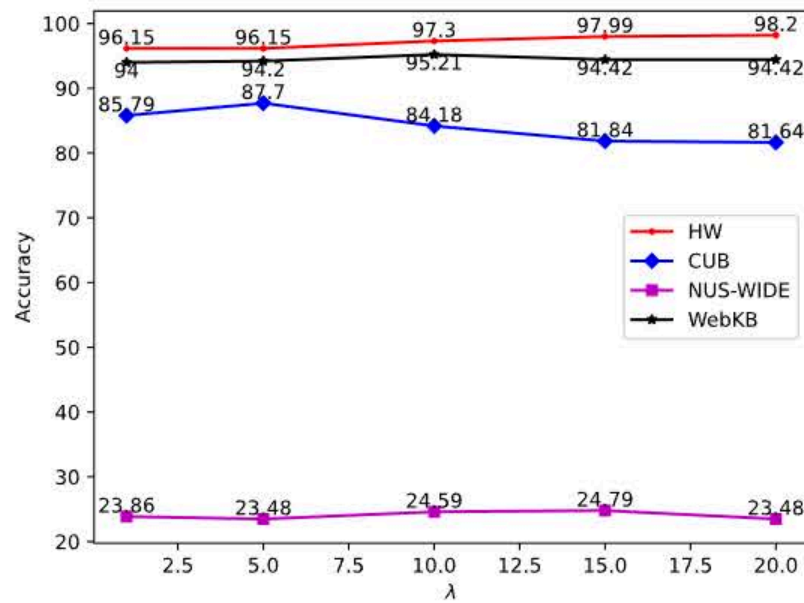


Fig. 5. Parameter tuning on HW, CUB, NUS-WIDE, and WebKB.

networks is much better than that of using a single model. The reason is that when we utilize a concatenated representation of view-specific as a global representation, multiple independent networks guarantee the capacity for complementary information to enhance the performance of classification.

3) *Is the Memory Bank Necessary?*: The memory bank is used to mitigate the confirmation bias caused by pseudo-labeling. To demonstrate its necessity, we conduct experiments and show the results of our model with/without memory bank in Table VI. It can be seen that the memory bank is usually very effective in improving classification performance.

D. Parameter Sensitivity

This experiment shows the parameter tuning with respect to λ on four datasets with $\sigma = 0.1$. Results are shown in Fig. 5, and we find that $\lambda = 15$ gives the best performance on NUS-WIDE, $\lambda = 10$ gives the best performance on WebKB. It indicates that the MMatch can achieve promising performance with suitable parameters. Besides, MMatch can achieve good performance on a wide range of parameters, indicating its robustness. And this phenomenon further demonstrates that the proposed method learns discriminative representation so that it is insensitive to λ .

VI. CONCLUSION

This work proposes a novel semi-supervised multi-view learning framework, which provides a new choice for unlabeled multi-view data. The proposed MMatch overcomes the adverse effects caused by weak discriminative views. Moreover, it reduces the difference between multiple views using a consistent semi-supervised classification. The learned representation contains information on consistency, complementarity, and strong discriminability.

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表 VASMIN (μ = 0.1)。

“SM” ISM				
Metrics	Method	HW	Caltech101-7	WebKB
ACC	ours	98.05	95.54	95.03
	SM	94.64	92.80	89.83
Precision	ours	98.11	81.75	94.04
	SM	94.96	69.30	92.44
Recall	ours	98.24	81.51	93.15
	SM	94.72	65.07	76.60
F1	ours	97.97	79.81	92.51
	SM	94.25	65.05	80.81

表六

ASMB (μ = 0.1) 。’’’’我
W /WMB, R

Metrics	Method	HW	Caltech101-7	WebKB
ACC	✓	98.05	95.54	95.03
	✗	97.77	94.14	94.89
Precision	✓	98.11	81.57	94.04
	✗	97.99	71.41	90.50
Recall	✓	98.24	81.51	93.51
	✗	97.79	69.76	79.20
F1	✓	97.97	79.81	92.51
	✗	97.65	68.50	80.72

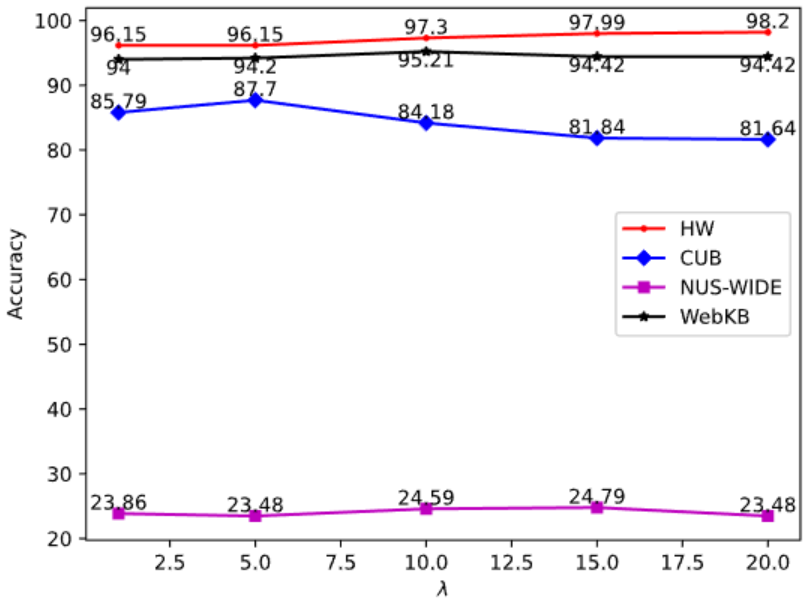


图 5。 对 HW、CUB、NUS-WIDE 和 WebKB 进行参数调优。

网络比使用单一模型要好得多。原因在于，当我们利用特定视角的级联表示作为全局表示时，多个独立的网络能够保证互补信息的利用，从而提升分类性能。

3) 记忆库是必要的吗?：记忆库用于缓解伪标签引起的确认偏差。为了证明其必要性，我们进行了实验，并在表 VI 中展示了使用/不使用记忆库的模型结果。可以看出，记忆库通常能非常有效地提高分类性能。

D. 参数灵敏度

本实验展示了在 μ=0.1 的条件下，针对四个数据集对参数 λ 进行调优的结果。结果如图 5 所示，我们发现 λ=15 在 NUS-WIDE 数据集上表现最佳，λ=10 在 WebKB 数据集上表现最佳。这表明，MMatch 算法在合适的参数下能够取得令人满意的性能。此外，MMatch 算法在较宽的参数范围内均能取得良好的性能，表明其具有良好的鲁棒性。这一现象进一步证明了所提出的方法学习到了判别性表示，因此对 λ 值不敏感。

VI. C 本文提出了一种新颖的半监督多视图学习框架，为处理无标签多视图数据提供了一种新的选择。所提出的 MMatch 框架克服了弱区分性视图带来的不利影响。此外，它利用一致的半监督分类方法降低了多个视图之间的差异。学习到的表示包含了一致性、互补性和强区分性的信息。

R

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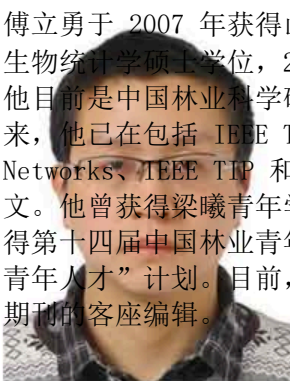
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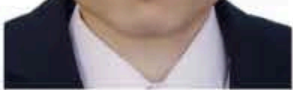
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